

Daniel Pitzele

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EDUCATION

University of Pennsylvania

Ph.D. in Computer Science

Philadelphia, PA

Aug. 2026 – Present

- Advised by Dr. Camillo J. Taylor in the GRASP laboratory
- GPA: N/A

University of Maryland

B.S. in Computer Science, Minor in Philosophy

College Park, MD

Aug. 2022 – May 2026

- GPA: 3.977
- Conducted undergraduate research under Dr. Jia-Bin Huang, Dr. Christopher Metzler, and as a part of the Gemstone Research Honors Program
- Relevant Coursework: Computer Vision, Robotics, Computational Photography, Multimodal Deep Learning, Machine Learning for Physical Sensing and Perception, Theory of Computation, Algorithms, etc.

RESEARCH/PROJECTS

Slit Aperture for Fourier Ptychography | *with Dr. Christopher Metzler*

April 2025 - July 2026

- Constructed and precisely aligned a Fourier ptychography setup on an optical bench
- Utilized a slit aperture as a proof of concept for spatially restricted systems
- Developed Fourier ptychography reconstruction code that works with a slit aperture
- Presenting a [co-first authored poster](#) at ICCP 2026 and created a [highlight video](#) for the project

Color Vision Simulation | *with Dr. Giuliano Scarcelli*

November 2023 - May 2026

- Worked with a team of undergraduates to develop a simulation to visualize what a person with CVD sees
- Conducted an IRB-approved human study to verify the efficacy of our simulation
- Orally presented and defended the team's undergraduate thesis, winning Gemstone's "Best Thesis Presentation" award among the senior teams
- Presented a poster at University of Maryland's Undergraduate Research Day 2025

Trajectory Guidance for Robot Policies | *with Dr. Jia-Bin Huang*

August 2025 - June 2026

- Compared different methods of trajectory guidance for robot policies
- Utilized both diffusion policy and VLA-based architectures to create a robust set of experiments
- Conducted extensive review of literature in the area of trajectory guidance

Autonomous Audio Navigation in Caves

October 2025 - December 2025

- Designed a machine learning model for autonomous navigation using only audio signals
- Created a cave generation and audio propagation pipeline for gathering training examples
- Performed quantitative evaluations of our model and produced a conference-style [write-up](#)

Geolocatability Classifier

October 2025 - December 2025

- Created a series of machine learning models to estimate the ability of large VLMs to geolocate an image
- Gathered a dataset of SOTA VLMs' attempts to guess the location of an image
- Evaluated various adversarial filters for anti-geolocation purposes
- Performed quantitative evaluations of our models and produced a conference-style [write-up](#)

Machine Learning Colorization Engine

May 2025

- Implemented a machine learning model for colorization of black-and-white images using a U-Net architecture
- Trained the model using a combination of various open-source datasets
- Calibrated various model versions to work in cases of landscapes, people, and general images

Exercise Games for Seniors

January 2024 - May 2024

- Developed a lower-body focused exercise game for seniors in Unity using C# as part of research under a professor
- Completed a literature review on the subject to determine good exercises and formats to use
- Designed and implemented a fitness video game with strict technological requirements and minimal oversight

- Utilized various computer vision techniques and libraries for pose estimation and tracking

Wikimedia Anti-Disinformation Map

August 2023 – February 2024

- Worked on a team of undergraduates to develop a React web app for a research project under a professor
- Parsed, analyzed, and visualized data sets representing hundreds of anti-disinformation projects
- Corresponded with both a professor and contacts at Wikimedia to edit the site as needed
- Used GitHub actions to asynchronously update and redeploy the website at will

WORK EXPERIENCE

Software Engineer Intern

June 2025 – August 2025

Capital One

McLean, VA

- Implemented core frontend features of a VPN split tunnel provisioning tool in ReactJS to improve user experience
- Developed a backend API using Node.js and FastAPI, integrating both internal and external APIs
- Collaborated with 3 other interns to build a full-stack solution leveraging AWS services (ECS, S3, Load Balancer), enhancing application performance and reliability in a production-grade environment

Software Engineer Intern

June 2024 – August 2024

Strata Decision Technology

Chicago, IL

- Created a frontend interface using Typescript and React to be used for a microservice deployed through AWS
- Developed a RESTful API backend using .NET, C#, and Postgres/SQL
- Integrated the new frontend and backend for full stack development using an Agile methodology

President

November 2022 – November 2023

Maryland Men's Crew

College Park, MD

- Directly managed practice, competition, logistics, and travel of a 30-person rowing team
- Served as a liaison between the team, the university, other local rowing organizations, and alumni
- Allocated and executed the spending of a \$25,000/year team budget

Human Resources Intern

June 2023 – August 2023

Titan Security

Chicago, IL

- Worked directly and indirectly with candidates and their documents
- Processed/entered data in Microsoft Excel and various other office platforms
- Assisted candidates with their application/onboarding process

Information Technology Intern

February 2022 - June 2022

Relativity

Chicago, IL

- Assisted users directly using critical thinking and problem-solving skills
- Handled the storage, distribution, and inventory of company devices
- Earned the Google IT Support Specialization Certification

TECHNICAL SKILLS

Languages: Python, Java, Matlab, JavaScript/Typescript, C#, C/C++, OCaml, Rust, HTML/CSS, SQL

Frameworks/Libraries: NumPy, PyTorch, React, OpenCV, Node.js, FastAPI, Material UI

Developer Tools: Git/GitHub, AWS, VS Code, Docker, Visual Studio, Octopus, Google Cloud Platform, GitKraken, Swagger

Misc. Technologies: LaTeX, Microsoft Word, Excel, PowerPoint